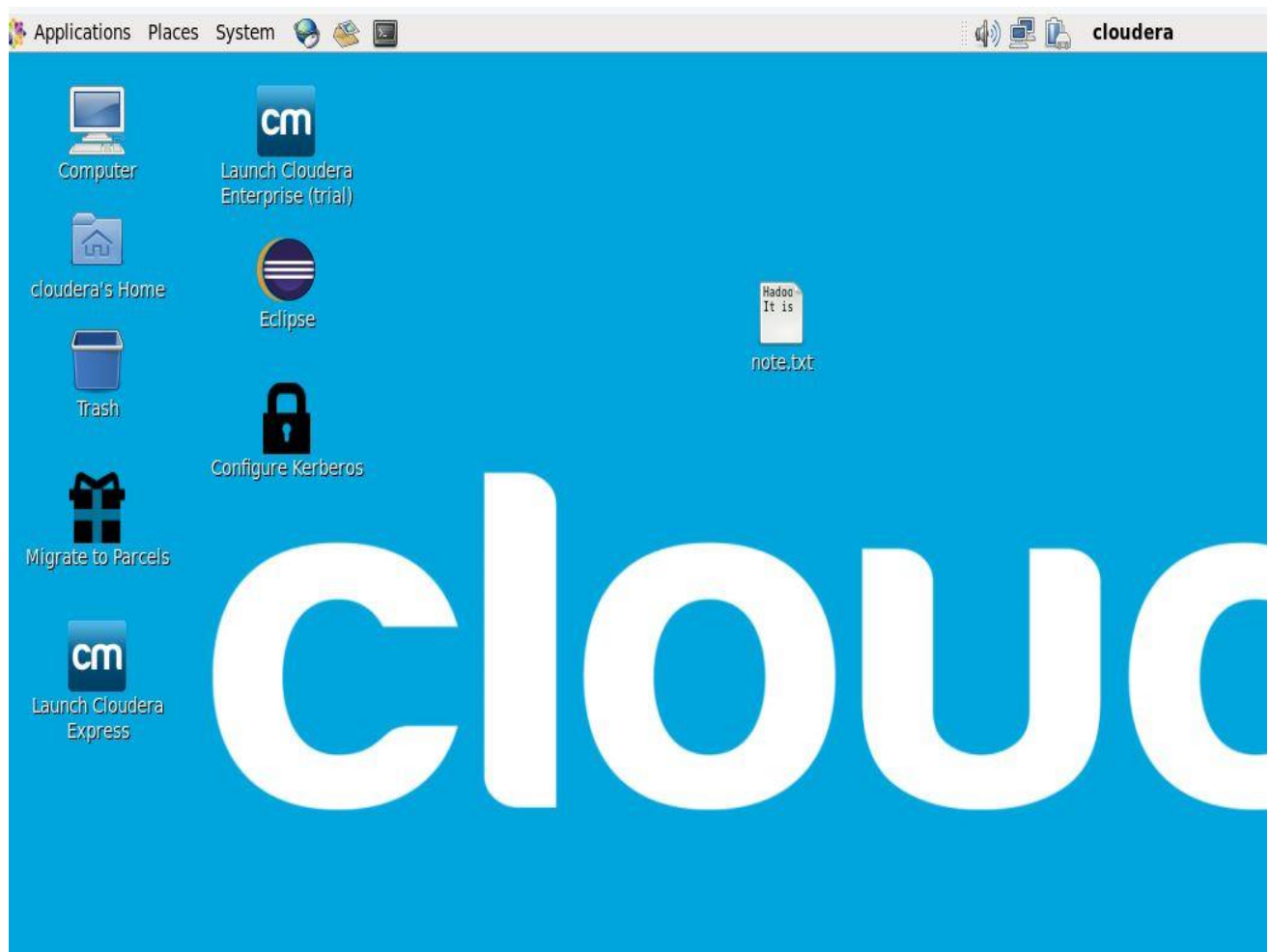


Task- interaction with hadoop file system commands

hadoop fs -

- 2. copy a file from cloudera desktop (or other place) to hadoop fs**
- 3. copy a file/folder from hadoop fs to cloudera home/any location**
- 4. checked how to run jar**
- 5. one jar file executed for mapreduce task for wordcount from a file .**
the mapreduce example was there in usr/lib/hadoop-mapreduce-examples.jar

Firstly I have created note.txt file as shown in below picture. And I typed some content in this file.

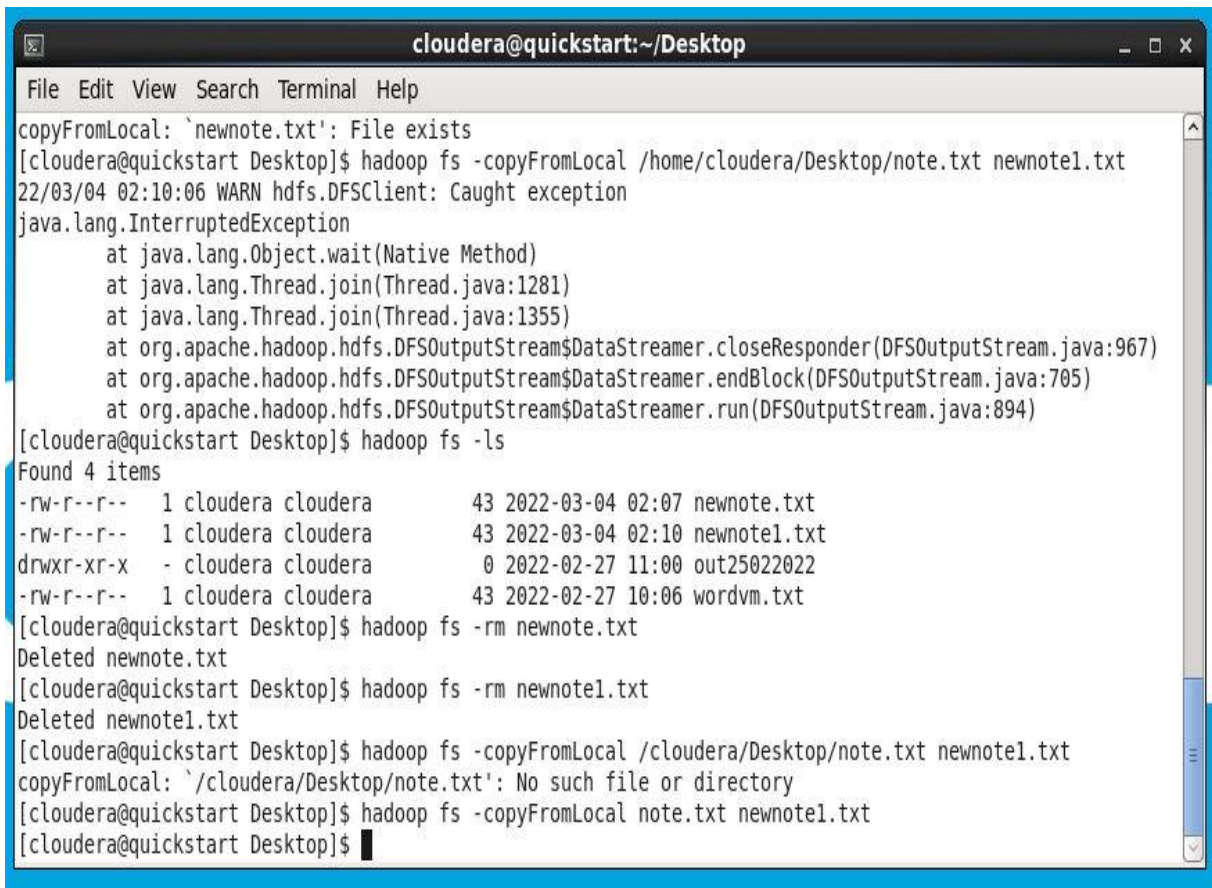


Hadoop File System

copy a file from cloudera desktop (or other place) to hadoop fs

To copy a file from your cloudera desktop to hadoop file system i used below command -:

hadoop fs -copyFromLocal note.txt newnote1.txt



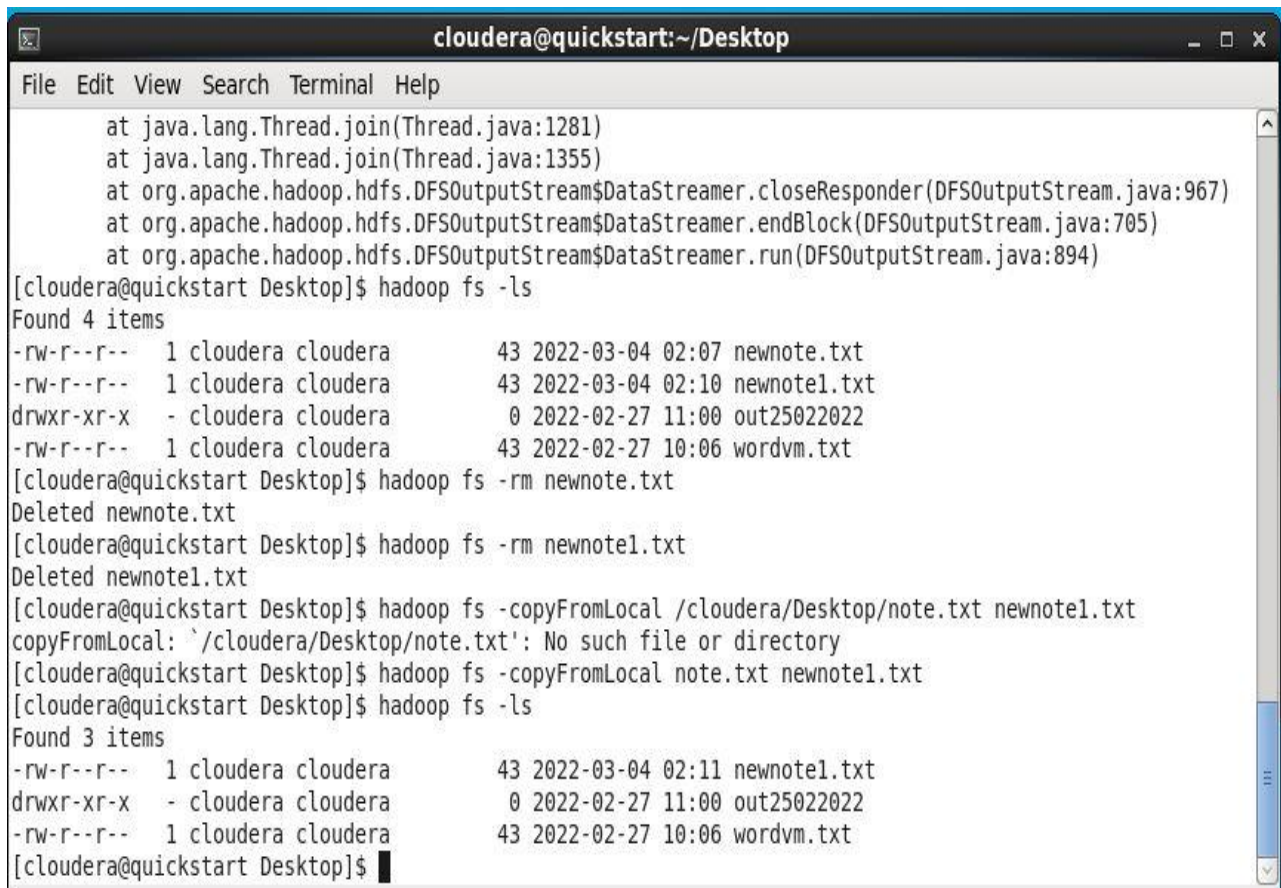
```
cloudera@quickstart:~/Desktop
File Edit View Search Terminal Help
copyFromLocal: `newnote.txt': File exists
[cloudera@quickstart Desktop]$ hadoop fs -copyFromLocal /home/cloudera/Desktop/note.txt newnote1.txt
22/03/04 02:10:06 WARN hdfs.DFSClient: Caught exception
java.lang.InterruptedException
    at java.lang.Object.wait(Native Method)
    at java.lang.Thread.join(Thread.java:1281)
    at java.lang.Thread.join(Thread.java:1355)
    at org.apache.hadoop.hdfs.DFSOutputStream$DataStreamer.closeResponder(DFSOutputStream.java:967)
    at org.apache.hadoop.hdfs.DFSOutputStream$DataStreamer.endBlock(DFSOutputStream.java:705)
    at org.apache.hadoop.hdfs.DFSOutputStream$DataStreamer.run(DFSOutputStream.java:894)
[cloudera@quickstart Desktop]$ hadoop fs -ls
Found 4 items
-rw-r--r--  1 cloudera cloudera      43 2022-03-04 02:07 newnote.txt
-rw-r--r--  1 cloudera cloudera      43 2022-03-04 02:10 newnote1.txt
drwxr-xr-x  - cloudera cloudera      0 2022-02-27 11:00 out25022022
-rw-r--r--  1 cloudera cloudera      43 2022-02-27 10:06 wordvm.txt
[cloudera@quickstart Desktop]$ hadoop fs -rm newnote.txt
Deleted newnote.txt
[cloudera@quickstart Desktop]$ hadoop fs -rm newnote1.txt
Deleted newnote1.txt
[cloudera@quickstart Desktop]$ hadoop fs -copyFromLocal /cloudera/Desktop/note.txt newnote1.txt
copyFromLocal: `/cloudera/Desktop/note.txt': No such file or directory
[cloudera@quickstart Desktop]$ hadoop fs -copyFromLocal note.txt newnote1.txt
[cloudera@quickstart Desktop]$
```

Hadoop File System

You can see newnote1.txt file is there in hadoop filesystem.

I have checked it through below command -:

hadoop fs -ls



```
cloudera@quickstart:~/Desktop
File Edit View Search Terminal Help
  at java.lang.Thread.join(Thread.java:1281)
  at java.lang.Thread.join(Thread.java:1355)
  at org.apache.hadoop.hdfs.DFSOutputStream$DataStreamer.closeResponder(DFSOutputStream.java:967)
  at org.apache.hadoop.hdfs.DFSOutputStream$DataStreamer.endBlock(DFSOutputStream.java:705)
  at org.apache.hadoop.hdfs.DFSOutputStream$DataStreamer.run(DFSOutputStream.java:894)
[cloudera@quickstart Desktop]$ hadoop fs -ls
Found 4 items
-rw-r--r--  1 cloudera cloudera      43 2022-03-04 02:07 newnote.txt
-rw-r--r--  1 cloudera cloudera      43 2022-03-04 02:10 newnote1.txt
drwxr-xr-x  - cloudera cloudera      0 2022-02-27 11:00 out25022022
-rw-r--r--  1 cloudera cloudera      43 2022-02-27 10:06 wordvm.txt
[cloudera@quickstart Desktop]$ hadoop fs -rm newnote.txt
Deleted newnote.txt
[cloudera@quickstart Desktop]$ hadoop fs -rm newnote1.txt
Deleted newnote1.txt
[cloudera@quickstart Desktop]$ hadoop fs -copyFromLocal /cloudera/Desktop/note.txt newnote1.txt
copyFromLocal: `/cloudera/Desktop/note.txt': No such file or directory
[cloudera@quickstart Desktop]$ hadoop fs -copyFromLocal note.txt newnote1.txt
[cloudera@quickstart Desktop]$ hadoop fs -ls
Found 3 items
-rw-r--r--  1 cloudera cloudera      43 2022-03-04 02:11 newnote1.txt
drwxr-xr-x  - cloudera cloudera      0 2022-02-27 11:00 out25022022
-rw-r--r--  1 cloudera cloudera      43 2022-02-27 10:06 wordvm.txt
[cloudera@quickstart Desktop]$
```

copy a file/folder from hadoop fs to cloudera home/any location

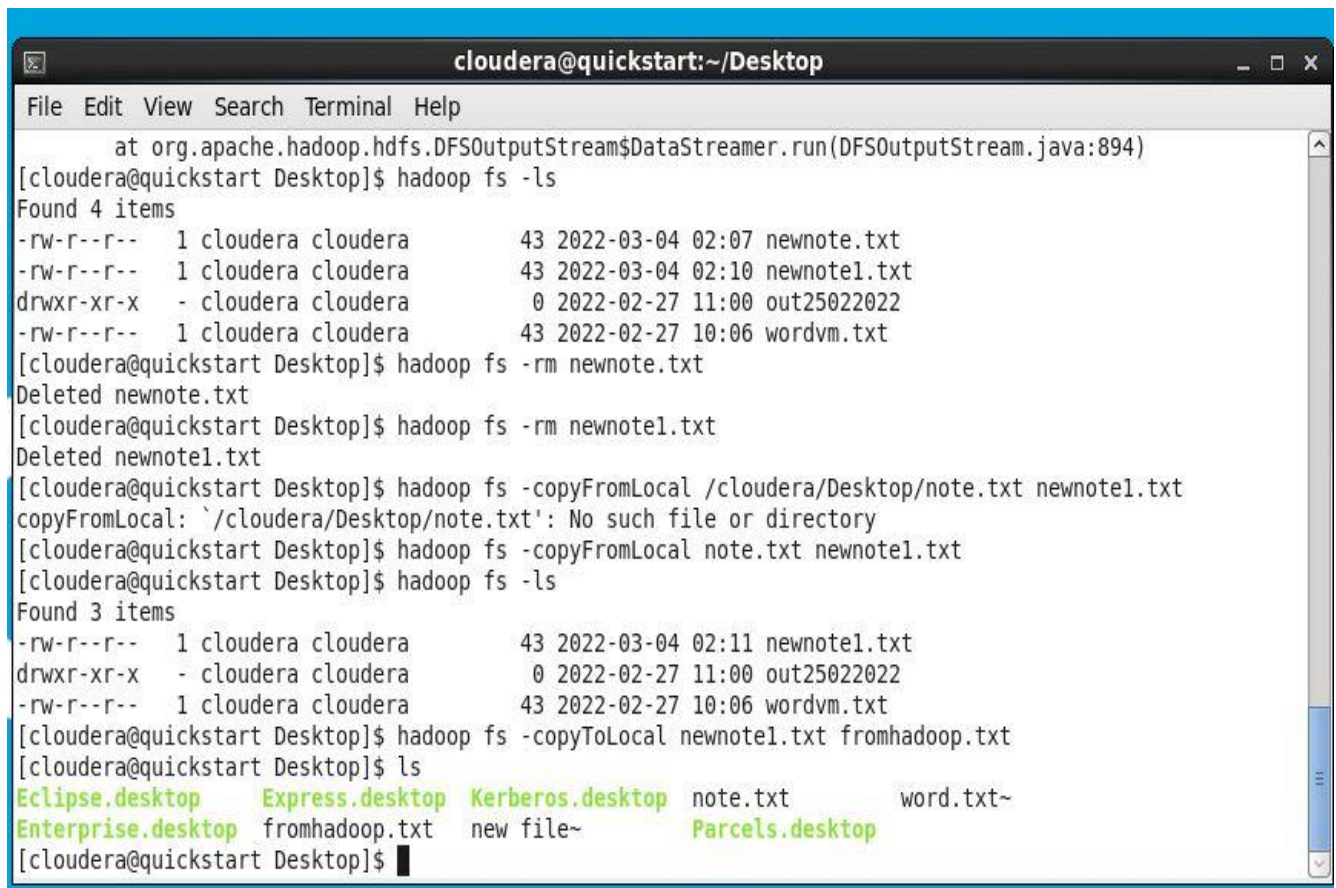
To copy the files from hadoop file system to my desktop I used the below command -:

hadoop fs -copyToLocal newnote1.txt fromhadoop.txt

After copying the file I checked my directory through below command -
:

Ls command

You can see fromhadoop.txt is there in my desktop directory.



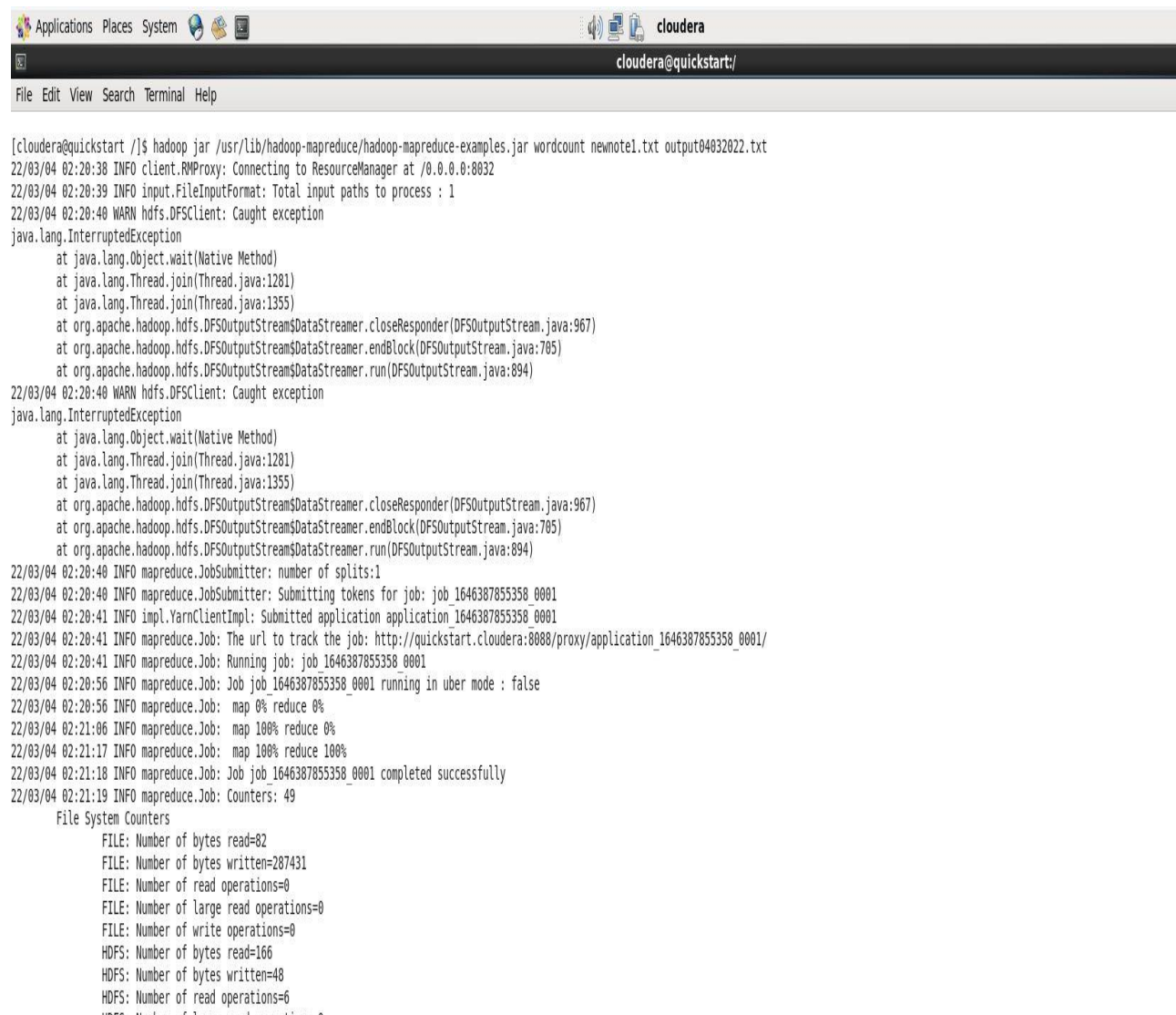
```
cloudera@quickstart:~/Desktop
File Edit View Search Terminal Help
at org.apache.hadoop.hdfs.DFSOutputStream$DataStreamer.run(DFSOutputStream.java:894)
[cloudera@quickstart Desktop]$ hadoop fs -ls
Found 4 items
-rw-r--r-- 1 cloudera cloudera 43 2022-03-04 02:07 newnote.txt
-rw-r--r-- 1 cloudera cloudera 43 2022-03-04 02:10 newnote1.txt
drwxr-xr-x - cloudera cloudera 0 2022-02-27 11:00 out25022022
-rw-r--r-- 1 cloudera cloudera 43 2022-02-27 10:06 wordvm.txt
[cloudera@quickstart Desktop]$ hadoop fs -rm newnote.txt
Deleted newnote.txt
[cloudera@quickstart Desktop]$ hadoop fs -rm newnote1.txt
Deleted newnote1.txt
[cloudera@quickstart Desktop]$ hadoop fs -copyFromLocal /cloudera/Desktop/note.txt newnote1.txt
copyFromLocal: `/cloudera/Desktop/note.txt': No such file or directory
[cloudera@quickstart Desktop]$ hadoop fs -copyFromLocal note.txt newnote1.txt
[cloudera@quickstart Desktop]$ hadoop fs -ls
Found 3 items
-rw-r--r-- 1 cloudera cloudera 43 2022-03-04 02:11 newnote1.txt
drwxr-xr-x - cloudera cloudera 0 2022-02-27 11:00 out25022022
-rw-r--r-- 1 cloudera cloudera 43 2022-02-27 10:06 wordvm.txt
[cloudera@quickstart Desktop]$ hadoop fs -copyToLocal newnote1.txt fromhadoop.txt
[cloudera@quickstart Desktop]$ ls
Eclipse.desktop  Express.desktop  Kerberos.desktop  note.txt          word.txt~
Enterprise.desktop  fromhadoop.txt  new file~          Parcels.desktop
[cloudera@quickstart Desktop]$
```

Hadoop File System

one jar file executed for mapreduce task for wordcount from a file . the mapreduce example was there in usr/lib/hadoop-mapreduce-examples.jar

To execute mapreduce task I used following command

hadoop jar /usr/lib/hadoop-mapreduce/hadoop-mapreduce-examples.jar newnote1.txt output04032022



```
[cloudera@quickstart /]$ hadoop jar /usr/lib/hadoop-mapreduce/hadoop-mapreduce-examples.jar wordcount newnote1.txt output04032022.txt
22/03/04 02:20:38 INFO client.RMProxy: Connecting to ResourceManager at /0.0.0.0:8032
22/03/04 02:20:39 INFO input.FileInputFormat: Total input paths to process : 1
22/03/04 02:20:40 WARN hdfs.DFSClient: Caught exception
java.lang.InterruptedException
    at java.lang.Object.wait(Native Method)
    at java.lang.Thread.join(Thread.java:1281)
    at java.lang.Thread.join(Thread.java:1355)
    at org.apache.hadoop.hdfs.DFSOutputStream$DataStreamer.closeResponder(DFSOutputStream.java:967)
    at org.apache.hadoop.hdfs.DFSOutputStream$DataStreamer.endBlock(DFSOutputStream.java:705)
    at org.apache.hadoop.hdfs.DFSOutputStream$DataStreamer.run(DFSOutputStream.java:894)
22/03/04 02:20:40 WARN hdfs.DFSClient: Caught exception
java.lang.InterruptedException
    at java.lang.Object.wait(Native Method)
    at java.lang.Thread.join(Thread.java:1281)
    at java.lang.Thread.join(Thread.java:1355)
    at org.apache.hadoop.hdfs.DFSOutputStream$DataStreamer.closeResponder(DFSOutputStream.java:967)
    at org.apache.hadoop.hdfs.DFSOutputStream$DataStreamer.endBlock(DFSOutputStream.java:705)
    at org.apache.hadoop.hdfs.DFSOutputStream$DataStreamer.run(DFSOutputStream.java:894)
22/03/04 02:20:40 INFO mapreduce.JobSubmitter: number of splits:1
22/03/04 02:20:40 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1646387855358_0001
22/03/04 02:20:41 INFO impl.YarnClientImpl: Submitted application application_1646387855358_0001
22/03/04 02:20:41 INFO mapreduce.Job: The url to track the job: http://quickstart.cloudera:8088/proxy/application_1646387855358_0001/
22/03/04 02:20:41 INFO mapreduce.Job: Running job: job_1646387855358_0001
22/03/04 02:20:56 INFO mapreduce.Job: Job job_1646387855358_0001 running in uber mode : false
22/03/04 02:20:56 INFO mapreduce.Job: map 0% reduce 0%
22/03/04 02:21:06 INFO mapreduce.Job: map 100% reduce 0%
22/03/04 02:21:17 INFO mapreduce.Job: map 100% reduce 100%
22/03/04 02:21:18 INFO mapreduce.Job: Job job_1646387855358_0001 completed successfully
22/03/04 02:21:19 INFO mapreduce.Job: Counters: 49
File System Counters
  FILE: Number of bytes read=82
  FILE: Number of bytes written=287431
  FILE: Number of read operations=0
  FILE: Number of large read operations=0
  FILE: Number of write operations=0
  HDFS: Number of bytes read=166
  HDFS: Number of bytes written=48
  HDFS: Number of read operations=6
  HDFS: Number of write operations=6
```

Hadoop File System

```
cloudera@quickstart:/
File Edit View Search Terminal Help
HDFS: Number of read operations=6
HDFS: Number of large read operations=0
HDFS: Number of write operations=2
Job Counters
  Launched map tasks=1
  Launched reduce tasks=1
  Data-local map tasks=1
  Total time spent by all maps in occupied slots (ms)=7836
  Total time spent by all reduces in occupied slots (ms)=8273
  Total time spent by all map tasks (ms)=7836
  Total time spent by all reduce tasks (ms)=8273
  Total vcore-milliseconds taken by all map tasks=7836
  Total vcore-milliseconds taken by all reduce tasks=8273
  Total megabyte-milliseconds taken by all map tasks=8024064
  Total megabyte-milliseconds taken by all reduce tasks=8471552
Map-Reduce Framework
  Map input records=3
  Map output records=9
  Map output bytes=78
  Map output materialized bytes=82
  Input split bytes=123
  Combine input records=9
  Combine output records=7
  Reduce input groups=7
  Reduce shuffle bytes=82
  Reduce input records=7
  Reduce output records=7
  Spilled Records=14
  Shuffled Maps =1
  Failed Shuffles=0
  Merged Map outputs=1
  GC time elapsed (ms)=220
  CPU time spent (ms)=1930
  Physical memory (bytes) snapshot=370278400
  Virtual memory (bytes) snapshot=3015192576
  Total committed heap usage (bytes)=226365440
Shuffle Errors
  BAD_ID=0
  CONNECTION=0
  IO_ERROR=0
  WRONG_LENGTH=0
  WRONG_MAP=0
  WRONG_REDUCE=0
File Input Format Counters
  Bytes Read=43
File Output Format Counters
  Bytes Written=48
```

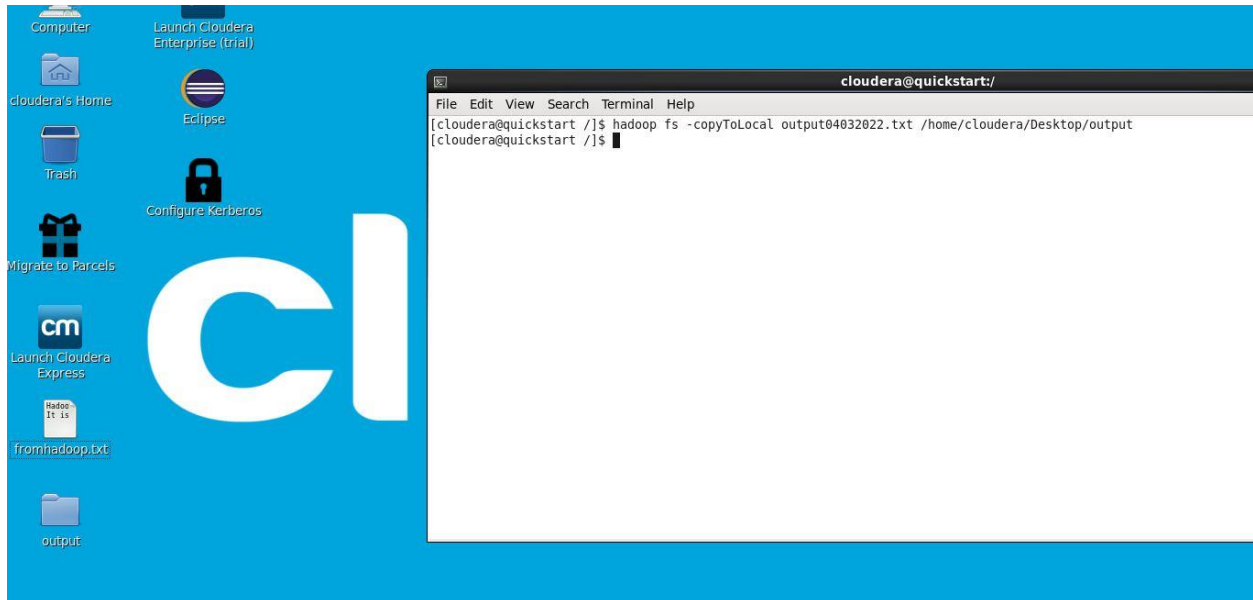
After executing mapreduce task you can see one file output04032022 is generated by map reduce task in hadoop file system. I have checked it through the following command:

Hadoop fs -ls

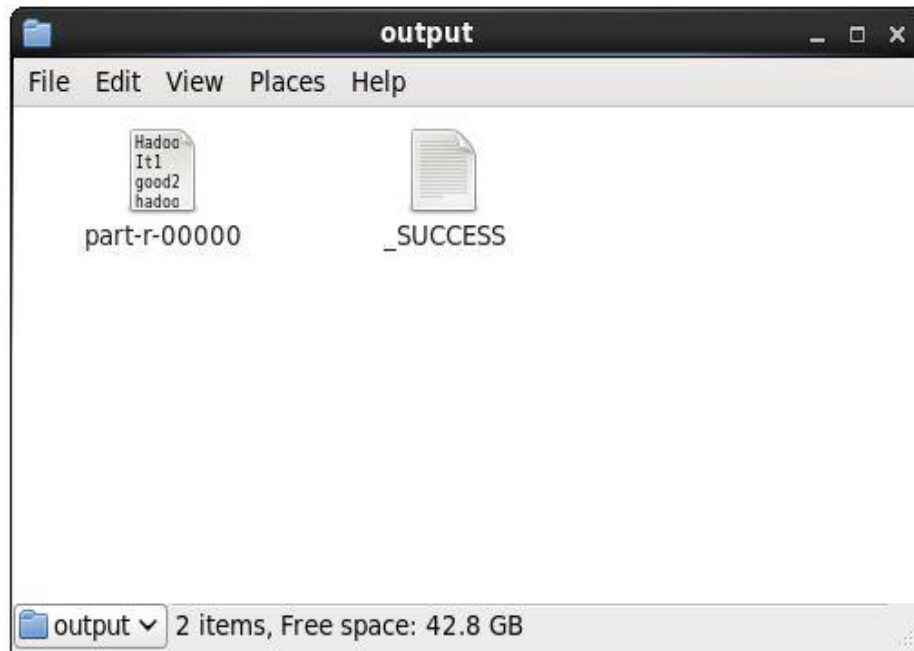
```
[cloudera@quickstart /]$ hadoop fs -ls
Found 4 items
-rw-r--r--  1 cloudera cloudera    43 2022-03-04 02:11 newnote1.txt
drwxr-xr-x  - cloudera cloudera     0 2022-02-27 11:00 out25022022
drwxr-xr-x  - cloudera cloudera     0 2022-03-04 02:21 output04032022.txt
-rw-r--r--  1 cloudera cloudera    43 2022-02-27 10:06 wordvm.txt
[cloudera@quickstart /]$
```

Hadoop File System

Then I just copy it from hadoop file system to my desktop.

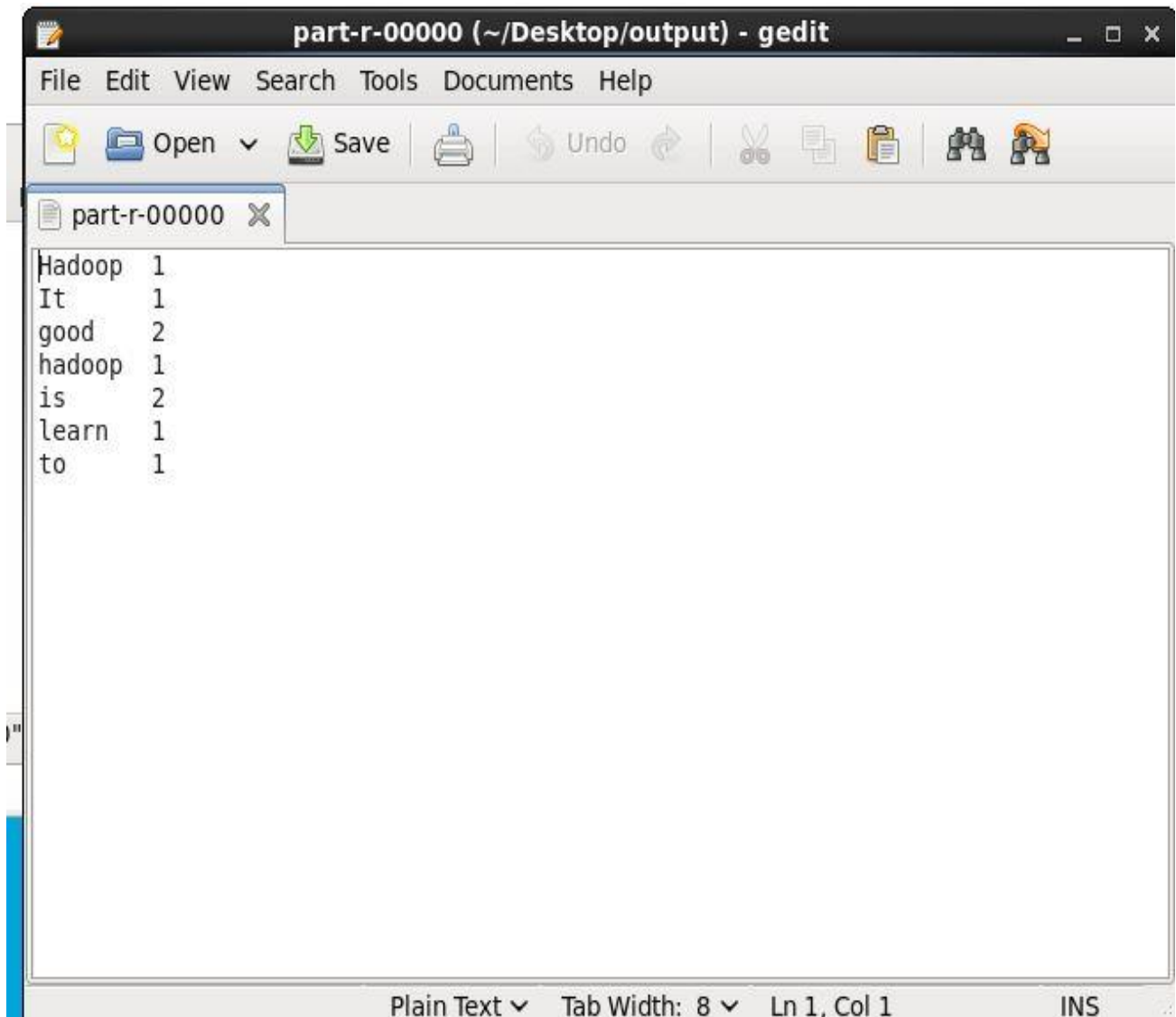


In output folder -:



Hadoop File System

In part-r-00000 file



The screenshot shows a gedit text editor window titled "part-r-00000 (~/Desktop/output) - gedit". The window has a menu bar with "File", "Edit", "View", "Search", "Tools", "Documents", and "Help". Below the menu bar is a toolbar with icons for "Open", "Save", "Print", "Undo", "Redo", "Cut", "Copy", "Paste", "Find", and "Run". The main text area contains the following text:

```
Hadoop 1
It 1
good 2
hadoop 1
is 2
learn 1
to 1
```

The status bar at the bottom of the window displays "Plain Text", "Tab Width: 8", "Ln 1, Col 1", and "INS".